Did you notice Batman's hesitation to use guns? That is because a gunman killed Wayne's parents.

Movies in 2009 100

A roundup of the best movies to look out for this year

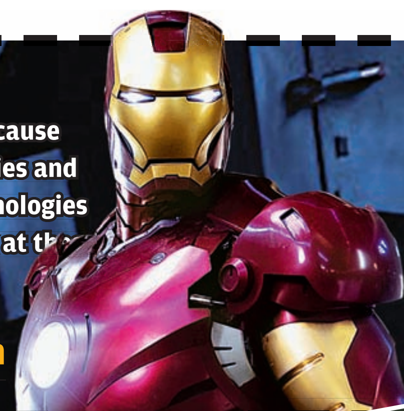
Batpod

The batpod detaches from the tumbler, and has a minimal structure for a bike. There are two engines inside the hubs of the wheels. The concept of the engine enclosed within the wheel has been around for a long time, but involves a complicated mechanism in real life that has to be figured out. The engines inside both the wheels will have to have microprocessors, and some sort of wireless or wired communication to be synchronised. An alternative would be to enclose the engine just in the back wheel of the motorcycle. The batpod has no seat, and Batman leans forward, almost lying with his stomach on the tank, with the back wheel between his legs. The Dodge Tomahawk requires a similar sitting position for riding, but is as yet only a concept. Moreover, the engine is located below the tank and not within the wheels. Because of the size of the tyres, and the top speed of the batpod, the entire upper body is used to steer the bike instead of the arms. Additionally, the light frame of the vehicle allows Batman to stop suddenly, and even go vertical if the situation calls for it. There have been a few concept cars that travel vertically on two wheels in city traffic, and horizontally on freeways, but these have not hit the streets yet.

Nokia 5800 XpressMusic (sans SONAR)

The most unscientific of all the tech depicted in *The Dark Knight*, was the conversion of mobile phones into SONAR devices. SONAR works by sending ultrasonic pulses that bounce off objects and return to the point of origin at different times and with different incident angles. These returned pulses are run through an algorithm that maps out the surroundings spatially. The 369-MHz ARM 11 processor on a Nokia 5800 might be impressive, but it simply could not process the amount of ultrasonic data that would be required, or display the input from another device, even if the connectivity issues were resolved. The mobile phone microphone or speakers just don't have the fidelity required to accurately map ultrasonic pulses. Converting all the mobile phones in a city into SONAR is even more problematic. Existing computer processing technologies simply cannot handle the workload, even if all the phones were remotely hacked to keep sending back data in high compression to reduce the bandwidth. However, the device itself is on the streets, and you can check it out in our Bazaar section.

Batman and Iron Man are two remarkable superheroes – mostly because they have no superpowers. Both are billionaires with large companies and a lot of money at their disposal – which is used for developing technologies to fight crime – a substitute for superpowers. We take a closer look at th fictional technologies, unearthing the real world parallels.



Iron Man

Arc Reactor

The Arc Reactor is a device that powers the factory of Stark Industries, and the various Iron Man suits. The technology behind the device is not explicitly mentioned in the movie, but a specification of 3 gigajoules per second is given – which is 3 GW of power. The Arc Reactor used for the second Iron Man suit had a capacity four times the original, which is more than any existing nuclear fission reactor can produce. Excluding exotic explanations such as anti-matter and space drive, the technology behind the Arc Reactor is almost certainly some form of nuclear fusion, which is the same kind of energy that drives the stars. A hypothetical device very similar to the arc reactor is the Tokamak, a magnetised torus that holds together highly energised plasma in isolation for energy generation. The Joint European Torus (JET) in the UK is the world's largest fusion reactor, and conducts experiments that have more massive implications for mankind than the LHC. The JET has produced short but powerful bursts of nuclear fusion energy. The reactor has the same torus shape as the Tokamak and the Arc Reactor installed at Stark Industries.

Iron Man Suit

The Iron Man suit is not just an armour with jets on the shoes. An underplayed aspect of the Iron Man suit is that it is an exoskeleton powered for augmenting tasks. Cyberdine, a Japanese company is already offering powered exoskeletons for hire. Called the Robot Suit HAL (Hybrid Assistive Limb), the suit gives the wearer superhuman strength. You can climb steep surfaces and lift heavy loads. Berkeley Bionics is developing ExoHiker, an attachment for legs that gathers and stores power from the movement of the legs. A company called Sarcos has developed what is arguably the best exoskeleton in existence. The XOS can lift extremely heavy weights, but looks a lot less bulkier than the competition. MIT's biomechatronics group, and Honda are also believed to be developing powered exoskeletons. Apart from military use, powered exoskeletons can also be used to replace wheelchairs for the handicapped.